SOFTWARE ENGINEERING PROJECT REPORT

NullWasted: Surplus Food Marketplace Platform with Agentic AI Assistant

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TABLE OF CONTENTS

CHAPTER I: INTRODUCTION..... 3
1.1 Project Definition and Aim..... 3
1.2 User Stories..... 4
1.4 Hardware and Software Requirements..... 5
1.5 Risk Analysis..... 7
CHAPTER II: SOFTWARE PROJECT MANAGEMENT PLAN..... 9
2.1 Organization and Team Roles..... 9
2.1.1 Team Structure and Domain Ownership..... 9
2.1.2 RACI Responsibility Matrix..... 9
2.2 Work Breakdown Structure (Tasks)..... 10
2.3 Project Schedule and Milestones..... 11
2.4 Risk Management and Mitigation..... 11
2.5 Project Management Tool Implementation: Trello..... 13
2.5.1 Board Setup and Workflow Stages..... 13
2.5.2 Architectural Labels..... 13
2.5.3 Card Structure Standard..... 14
CHAPTER III: SYSTEM ARCHITECTURE AND DESIGN..... 15
3.1 High-Level Architecture (The Helicopter View)..... 15
3.1.1 Architectural Philosophy and Multi-Tier Topology..... 15
3.1.2 High-Level Architecture Diagram..... 16
3.1.3 Subsystem Responsibilities and Data Flow..... 17
3.1.4 Architectural Invariants and Trade-Off Rationale..... 17
3.2 Unified Modeling Language (UML) Diagrams..... 18
3.2.1 Use Case Diagrams (Functional View)..... 18
3.2.2 Activity Diagrams (Process and Behavioral View)..... 19
3.2.3 Class Diagrams (Structural Domain Model)..... 22
3.2.4 Sequence Diagrams (Dynamic Interaction View)..... 24
3.2.5 State Machine Diagrams (Lifecycle Statechart View)..... 24
3.2.6 Deployment Diagram (Physical and Container Topology)..... 26
3.3 Graphical User Interface (GUI) Design..... 27
3.3.1 Design Guidelines and Design System Tokens..... 27
3.3.2 Core Screen Mockups and User Flow Specifications..... 28

CHAPTER I: INTRODUCTION

Food waste represents one of the most pressing environmental and economic challenges in modern urban centers. Globally, roughly one third of all food produced for human consumption is lost or discarded each year. When organic matter decomposes in municipal landfills, it generates substantial volumes of methane gas, which has a global warming potential over twenty-five times higher than carbon dioxide. This ongoing waste pattern conflicts directly with sustainable development priorities, particularly responsible consumption and climate action.

In the commercial food and beverage sector, businesses such as bakeries, cafes, restaurants, and grocery retailers face regular inventory surpluses at the end of each business day. Even though shop owners understand that selling surplus meals at a discount recovers raw material costs, conventional processes create heavy friction. During closing hours, retail staff are occupied with kitchen cleaning, inventory counting, and cash settlement. Existing surplus platforms force staff to take photos, fill in descriptive forms, and manually set prices. Because of this administrative burden, throwing unsold food away remains the easiest path for tired staff.

To solve this operational barrier, **NullWasted** introduces an autonomous, AI-driven surplus food marketplace that enables exhausted store employees to list surplus inventory in seconds while providing budget-conscious consumers access to quality, affordable meals.

1.1 Project Definition and Aim

The primary aim of **NullWasted** is to design, develop, and deploy an accessible surplus food marketplace platform connecting local food merchants with consumers through scheduled self-pickup windows at discounts of fifty to seventy percent.

The core innovation of NullWasted lies in its autonomous Agentic AI store assistant, which eliminates manual administrative tasks through four primary functions:

1. **Natural Language Listing:** Staff report remaining items using simple conversational text or in-browser voice notes (via Web Audio API). The agent autonomously parses product attributes, categories, and quantities to create live catalog entries.
2. **Smart Bundling:** The agent groups scattered miscellaneous leftovers into appealing, discounted mystery bags, enabling stores to clear small portions without listing each item individually.
3. **Dynamic Pricing Recommendations:** The agent calculates optimal discount rates (50% to 70%) based on remaining shelf life, stock volume, and time remaining before store closure, which are permanently locked upon publication to prevent bargaining disputes.
4. **Food Safety Guardrail:** The agent checks items against BPOM regulatory safety thresholds via semantic vector similarity, automatically flagging or rejecting high-risk perishable items to maintain quality and protect consumer health.

Fulfillment follows a scheduled self-pickup model where customers collect their orders during a specific closing window, eliminating delivery overhead and keeping prices as low as possible.

1.2 User Stories

Stakeholder	User Story	Acceptance Criteria
Merchant Staff	As a store employee closing the shop, I want to report leftover food via short chat messages, so that items are listed without filling manual forms.	The AI extracts item names and counts, generating a preview card that staff can publish with a single tap.
Store Manager	As a store manager with miscellaneous leftover items, I want the system to package assorted foods into a discounted mystery bag, so that small portions can be sold together quickly.	The AI clusters small items into a themed mystery package, applies a 50% to 70% discount, and generates an appetizing description.
Store Cashier	As a store cashier, I want to verify customer orders by scanning a digital QR code, so that food handover is fast and recorded accurately.	Scanning validates the order in real time, marks it as completed in the database, and prevents duplicate redemptions.
Consumer	As a budget-conscious consumer, I want to browse discounted surplus meals and mystery bags near my location, so that I can buy quality food at an affordable price.	The feed displays store distance, discount percentage, remaining portions, and the exact pickup time window.
Consumer	As a customer, I want to reserve and pay for my surplus order online, so that my package is guaranteed and ready when I arrive at the store.	The platform processes digital payments via e-wallets or QRIS and generates a unique digital pickup pass.
Platform Admin	As a platform administrator, I want to monitor safety warnings and merchant compliance records, so that food safety standards are strictly enforced.	The system logs flagged high-risk items and requires supervisor review if a rejected listing is contested.

1.3 Similar Competitors

Evaluation Factor	NullWasted	Surplus Indonesia	Too Good To Go	Quick Commerce Apps
Primary Concept	AI-assisted surplus food marketplace with self-pickup	Manual surplus food rescue application	Global surplus mystery bag platform	On-demand commercial food delivery
Merchant Listing Effort	Conversational AI agent requiring minimal input (<30 seconds)	Manual form submission for each product	Manual daily inventory allocation	Detailed merchant portal with full menus
Product Packaging	Automated mystery bundles and single items	Individual discounted items and occasional packs	Mystery bags only	Full individual menu items
Pricing Mechanism	Dynamic AI suggestions between 50% and 70% off	Fixed merchant-chosen discounts	Fixed flat-rate mystery bag pricing	Standard retail pricing with promotional coupons
Food Safety Screening	Automated AI validation of ingredients and shelf life	Manual review and post-publication reports	Standard vendor compliance terms	General restaurant licensing
Fulfillment Method	Scheduled customer self-pickup	Manual review and post-publication reports	Scheduled customer self-pickup	Dedicated motorcycle couriers
Operational Effort for Staff	Under 30 seconds per closing shift	Several minutes of manual form entry	Low to moderate daily inventory checks	High overhead requiring active tablet management

1.4 Hardware and Software Requirements

Hardware Specifications

Component	Target User / Layer	Minimum Specification	Purpose
Merchant Device	F&B Store Staff	Android or iOS smartphone with camera and modern browser (Chrome/Safari), or POS terminal browser	Receiving chat orders, sending in-app voice notes via Web Audio API, and scanning customer QR vouchers via device camera

Component	Target User / Layer	Minimum Specification	Purpose
Consumer Device	General Public / Students	Modern smartphone running Android 10+ or iOS 14+ with GPS and mobile browser	Discovering nearby stores via Geolocation API, reserving surplus packages, and displaying digital QR pickup passes
Server Infrastructure	24/7 Container Host	Virtual machine or host with 2 vCPUs, 4 GB RAM, and 20 GB SSD storage	Hosting Docker microservices (FastAPI, 9Router AI Gateway, PostgreSQL + pgvector, Redis, Cloudflare Tunnel) with an idle footprint of ~350 MB RAM

Software Architecture and Technology Stack

Layer	Technology	Selection Rationale
Frontend Application	Responsive Web App & PWA (React 18/19, Vite, TypeScript, Tailwind CSS)	Delivers a lightweight, app-like mobile experience with native Web Audio recording, camera scanner, and Geolocation API without app store installation friction.
Backend API Server	Python 3.11+ with FastAPI & SQLModel	High-throughput asynchronous REST endpoints, strict Pydantic v2 data validation schemas, and native integration with the Python AI agent ecosystem.
Agentic AI & LLM Routing	LangGraph StateGraph & 9Router AI Gateway (Gemini Flash, Zhipu GLM-4, Groq Llama 3.3 & Whisper Large V3)	Stateful multi-step agent orchestration with automated multi-provider failover, sub-second Groq Whisper audio transcription, and deterministic schema enforcement.

Layer	Technology	Selection Rationale
Database & Vector Store	PostgreSQL 16 with pgvector & Redis 7 Alpine	PostgreSQL handles ACID transactional data and semantic BPOM embeddings, while Redis manages distributed stock reservation locks and closing-window countdown timers.
Infrastructure & Security	Docker Compose & Cloudflare Tunnel (`cloudflared`)	Reproducible multi-container orchestration with zero-trust encrypted public ingress, eliminating open firewall ports and DDoS risks.

1.5 Risk Analysis

Risk Description	Impact	Mitigation Strategy
Perishable food spoilage causing food safety issues for consumers	High	Enforce automated pgvector semantic similarity checks against BPOM food safety regulations, require preparation timestamps, and display reheating guidelines.
AI extraction errors misinterpreting items or quantities from staff messages	Medium	Enforce strict Pydantic schema validation on model outputs and present a 1-Tap interactive preview card for quick merchant verification prior to publishing.
Customer absence during the scheduled store pickup window	Medium	Require non-refundable upfront digital payment upon reservation and enforce strict 15-minute pickup countdown timers on digital QR vouchers.
Merchant staff neglecting surplus listings during busy closing shifts	High	Support in-app voice notes via Web Audio API (transcribed in <1s via Groq Whisper) and quick text notes to keep staff input under thirty seconds without third-party chat overhead.

Risk Description	Impact	Mitigation Strategy
Service latency during peak closing hours between eight and ten at night	Low	Deploy 9Router 3-tier fallback with asynchronous Redis task queues, token bucket rate limiting, and offline-capable PWA client caching.

CHAPTER II: SOFTWARE PROJECT MANAGEMENT PLAN

The Software Project Management Plan (SPMP) defines the governance, task allocation, master schedule, risk controls, and digital tracking mechanisms for **NullWasted** in accordance with IEEE Std 1058 and Eric J. Braude & Michael E. Bernstein (2010).

2.1 Organization and Team Roles

2.1.1 Team Structure and Domain Ownership

Group 16 operates as a Horizontal Agile Team with distributed domain ownership:

- **Vincenzo Fabian Tisila (2406354215) — Project Lead & AI Orchestration Lead**
  - **Responsibilities:** LangGraph agent workflow, 9Router AI gateway container, Groq Whisper STT pipeline, dynamic pricing algorithms, and smart mystery bag bundling.
  - **Deliverables:** Project charter, SPMP document, system architecture records, and sprint velocity tracking.
- **Michael Christian (2406348944) — Frontend UX & PWA Lead**
  - **Responsibilities:** React 18/19 PWA client, browser Web Audio recording interface, Geolocation store discovery feed, QR code voucher generation, and merchant camera QR scanner.
  - **Deliverables:** Frontend specifications, UI component library, client test plans, and usability audits.
- **Qais Ismail (2406487090) — Backend API & Database Infrastructure Lead**
  - **Responsibilities:** PostgreSQL 16 schema, pgvector store for BPOM food safety embeddings, Redis 7 atomic locking, SQLModel/Alembic migrations, FastAPI endpoints, and Cloudflare Tunnel ingress.
  - **Deliverables:** Database schema specifications, API contracts, container configurations, and CI/CD pipelines.

2.1.2 RACI Responsibility Matrix

Baseline Feature / Work Package	Vincenzo (AI)	Michael (FE)	Qais (BE)
FEAT-01: Natural Language Listing Pipeline	**A / R**	R	C
FEAT-02: Deterministic BPOM Safety Guardrail	C	I	**A / R**
FEAT-03: Smart Mystery Bag Bundling Engine	**A / R**	I	C

Baseline Feature / Work Package	Vincenzo (AI)	Michael (FE)	Qais (BE)
FEAT-04: Dynamic Pricing Recommendation	**A / R**	I	C
FEAT-05: 1-Tap Interactive Preview Card	C	**A / R**	I
FEAT-06: Store Active Listing CRUD	I	R	**A / R**
FEAT-07: Consumer Discovery Feed & Countdown	I	**A / R**	C
FEAT-08: Stock Reservation & Atomic Checkout	I	R	**A / R**
FEAT-09: Digital Pickup Pass with Unique QR	I	**A / R**	C
FEAT-10: Cashier Handover QR Scanner	I	**A / R**	R
INFRA: 9Router Gateway & Local Docker Stack	**A / R**	I	R
QA: Automated Testing & Verification Dossier	C	**A / R**	R

(Legend: A = Accountable; R = Responsible; C = Consulted; I = Informed).

2.2 Work Breakdown Structure (Tasks)

The engineering scope is broken down into 5 main packages and 18 deliverables:

- **1.0 Project Foundations & Lifecycle Governance**
  - 1.1 Inception & Requirements Specification (`DOC-01`)
  - 1.2 AI Architecture Canvas & Station Blueprint (`ARCH-01`)
  - 1.3 Software Project Management Plan & Tooling Setup (`DOC-02`)
  - 1.4 Quality Assurance & ISO/IEC 25010 Test Plan (`QA-01`)
  - 1.5 Final Operations Dossier, Release Notes & SBOM (`OPS-01`)
- **2.0 Core Infrastructure & Data Persistence**

    - 2.1 9Router Multi-Provider AI Gateway Container (`INFRA-01`)
    - 2.2 PostgreSQL 16 + pgvector & Redis Docker Stack (`DB-01`)
    - 2.3 Cloudflare Tunnel Secure Public Ingress (`INFRA-02`)
- **3.0 Agentic AI Engine & Food Safety Guardrails**
    - 3.1 Voice & Text Natural Language Listing Pipeline (`FEAT-01`)
    - 3.2 Deterministic BPOM Semantic Vector Guardrail (`FEAT-02`)
    - 3.3 Smart Bundling Mystery Bag Engine (`FEAT-03`)
    - 3.4 Dynamic Closing Window Pricing Engine (`FEAT-04`)
- **4.0 Merchant Portal Web Application**
    - 4.1 1-Tap Interactive Preview & Confirmation Card (`FEAT-05`)
    - 4.2 Active Catalog Management & Inventory CRUD (`FEAT-06`)
    - 4.3 In-App Camera QR Code Redemption Scanner (`FEAT-10`)
- **5.0 Consumer Marketplace & Pickup Flow**
    - 5.1 Geolocation Proximity Feed & Countdown Timers (`FEAT-07`)
    - 5.2 Stock Reservation & Atomic Checkout Module (`FEAT-08`)
    - 5.3 Offline-Capable Digital Pickup Pass with Signed QR (`FEAT-09`)

2.3 Project Schedule and Milestones

Development is executed over 16 academic weeks across four delivery sprints:

- **Phase 1: Inception & Architecture (Weeks 1–3) [Completed]**
    - Requirements specification (`DOC-01`) and AI Architecture Canvas (`ARCH-01`).
- **Phase 2: Sprint 1 — Merchant AI Pipeline & Core Infrastructure (Weeks 4–7) [In Progress]**
    - Delivery of SPMP (`DOC-02`), Docker stack setup (`DB-01`), 9Router gateway (`INFRA-01`), Voice Listing (`FEAT-01`), BPOM guardrail (`FEAT-02`), Dynamic pricing (`FEAT-04`), Preview card (`FEAT-05`), and Store CRUD (`FEAT-06`).
    - *Milestone: Midterm Examination (UTS) Live Demonstration.*
- **Phase 3: Sprint 2 — Consumer Marketplace & Bundling (Weeks 8–10)**
    - Mystery bag engine (`FEAT-03`), Geolocation discovery feed (`FEAT-07`), and Atomic stock reservation (`FEAT-08`).
- **Phase 4: Sprint 3 — Digital Voucher & QR Handover (Weeks 11–12)**
    - Customer QR pass (`FEAT-09`), Cashier camera QR scanner (`FEAT-10`), and Cloudflare Tunnel public ingress (`INFRA-02`).
- **Phase 5: Sprint 4 — Verification, Quality Audit & Release (Weeks 13–16)**
    - Automated testing suite (`QA-01`), SBOM, operations runbook, and Final Project Handover Dossier (`OPS-01`).
    - *Milestone: Final Examination (UAS) Production Demonstration.*

2.4 Risk Management and Mitigation

Risks are quantified using Eric J. Braude's formula: *Priority = (L × I) / C*, scored on a 1–10 scale:

Risk Domain	L	I	C	Priority	Mitigation Strategy
LLM API Quota Limits	7	8	3	**18.7**	Deploy 9Router with 3-tier fallback (Gemini Flash → GLM-4 → Groq).
AI Hallucinations	6	9	4	**13.5**	Enforce Pydantic validation schemas and mandatory 1-Tap preview review.
Perishable Food Spoilage	4	10	3	**13.3**	pgvector semantic filter auto-rejecting dairy/coconut milk > 4h at room temp.
Closing Shift Staff Rush	7	7	4	**12.3**	In-browser Web Audio voice notes transcribed via Groq Whisper in <1s.
Inventory Race Conditions	5	8	4	**10.0**	Redis distributed locking with 15-minute reservation TTLs for atomic deduction.
Camera Scan in Dim Light	6	6	4	**9.0**	HTML5 torch toggle support and 6-digit alphanumeric manual entry fallback.

Risk Domain	L	I	C	Priority	Mitigation Strategy
Timeline Scope Creep	6	7	5	**8.4**	Freeze V1 scope to 10 features; postpone reviews and CO2 trackers to V2.

2.5 Project Management Tool Implementation: Trello

2.5.1 Board Setup and Workflow Stages

The official NullWasted project management board is published at:

https://trello.com/b/6abf58e74fe44d966cceaa9c *(Title: NullWasted - Group 16 RPL)*

Work packages progress through five Kanban columns with strict Work-In-Progress (WIP) limits:

1. **Product Backlog:** All future sprint deliverables and backlog items.
2. **Sprint Backlog (Sprint 1):** Active sprint work packages committed for current delivery.
3. **In Progress:** Current active development tasks (WIP limit: max 3).
4. **Review and Validation:** Peer review, lint verification, and testing gate (WIP limit: max 2).
5. **Done:** Completed, tested, and verified work packages.

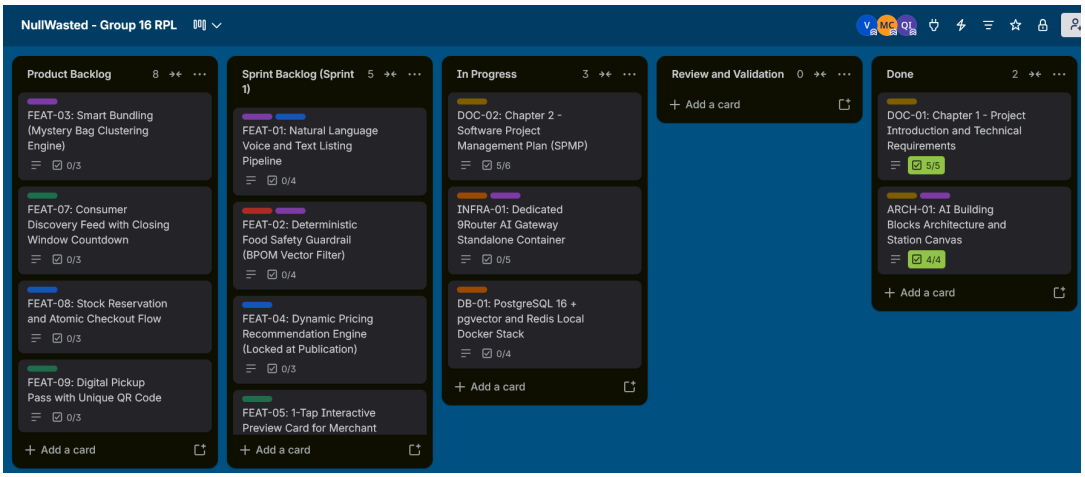


Figure 2.1: NullWasted Trello Kanban Board Setup and 5-Stage Workflow Pipeline.

2.5.2 Architectural Labels

Cards are categorized using six standardized color labels:

- **AI Agent Core (Purple):** LangGraph agent, 9Router, LLM prompts, and Whisper STT.
- **Backend & API (Blue):** FastAPI endpoints, pricing logic, and server algorithms.

- **Frontend PWA (Green):** React components, Web Audio recording, Geolocation, and QR.
- **Food Safety BPOM (Red):** Regulatory guardrails, vector embeddings, and safety rules.
- **Database & Infra (Orange):** Docker stacks, PostgreSQL/pgvector, Redis, and Cloudflare Tunnel.
- **Documentation & QA (Yellow):** Academic reports, SPMP documents, and test suites.

2.5.3 Card Structure Standard

Every card on the board follows a standardized layout:

- **Card Title:** `[ID] Name` (e.g., `FEAT-01: Natural Language Voice and Text Listing Pipeline`)
- **Labels:** `AI Agent Core`, `Frontend PWA`
- **Assignees:** Vincenzo Fabian Tisila (Lead), Michael Christian (Frontend Support)
- **Description:** Feature description linking functionality to architecture and user stories.
- **Acceptance Criteria Checklist:**
  - [ ] Web Audio API recording component in PWA
  - [ ] Groq Whisper Large V3 transcription pipeline (<1s)
  - [ ] LangGraph extraction agent with structured Pydantic schema
  - [ ] Manual text input fallback option

CHAPTER III: SYSTEM ARCHITECTURE AND DESIGN

Building upon the requirements baseline established in Chapter I and the Software Project Management Plan (SPMP) detailed in Chapter II, Chapter III formulates the architectural blueprints and interface specifications for **NullWasted**.

The architecture adheres to the international engineering standards ISO/IEC/IEEE 12207:2017 (*Software life-cycle processes*), the Object Management Group UML 2.5.1 specification, and the design principles outlined by Eric J. Braude & Michael E. Bernstein (2010).

3.1 High-Level Architecture (The Helicopter View)

3.1.1 Architectural Philosophy and Multi-Tier Topology

NullWasted is architected as an edge-oriented, cloud-native marketplace supported by a resilient agentic intelligence tier. Traditional surplus food platforms impose excessive administrative friction on retail store personnel, forcing tired workers during closing hours to manually type product names, navigate categories, upload high-resolution images, and set discounts.

To eliminate this operational friction, NullWasted adopts a 4-Tier distributed architecture where AI acts as an autonomous intermediary between unstructured human voice input and structured relational persistence:

1. **Client & Presentation Tier (Edge Devices):**
   - Responsive Progressive Web Application (PWA) built with React, Vite, and TypeScript.
   - Utilizes native browser Web Audio API for client-side PCM audio capture, HTML5 Geolocation API for store discovery, and HTML5 Canvas/MediaStream for cashier QR barcode scanning.
   - Runs natively on mobile browsers without requiring heavy app store downloads.
2. **Ingress, Network Security & Gateway Tier:**
   - Cloudflare Tunnel (`cloudflared`) providing end-to-end encrypted TLS ingress without opening public firewall ports.
   - Handles CORS enforcement, edge DDoS mitigation, and SSL/TLS termination.
3. **Application & Agentic Intelligence Tier (Compute Nodes):**
   - **Core REST API Service:** Asynchronous Python FastAPI framework implementing strict Pydantic v2 schemas and SQLModel ORM.
   - **Agentic AI Orchestrator:** LangGraph StateGraph engine coordinating voice transcription, natural language extraction, food safety evaluation, and dynamic pricing.
   - **Dedicated AI Gateway (9Router):** Standalone reverse proxy container managing multi-provider LLM failover across Google Gemini Flash, Zhipu GLM-4-Flash, and Groq Cloud (Llama 3.3 & Whisper Large V3).
4. **Persistence, Vector Store & Caching Tier:**

- **PostgreSQL 16 with pgvector:** Single source of truth for relational ACID data and semantic vector embeddings (BPOM safety regulations and catalog similarity).
- **Redis 7 Alpine:** High-speed in-memory store for distributed reservation locks, closing-window countdown timers, and rate-limiting buckets.

3.1.2 High-Level Architecture Diagram

The macro-level ("Helicopter View") topology illustrates the complete ecosystem and data boundaries between external human actors, edge client tiers, internal containerized microservices, and external LLM provider clusters:

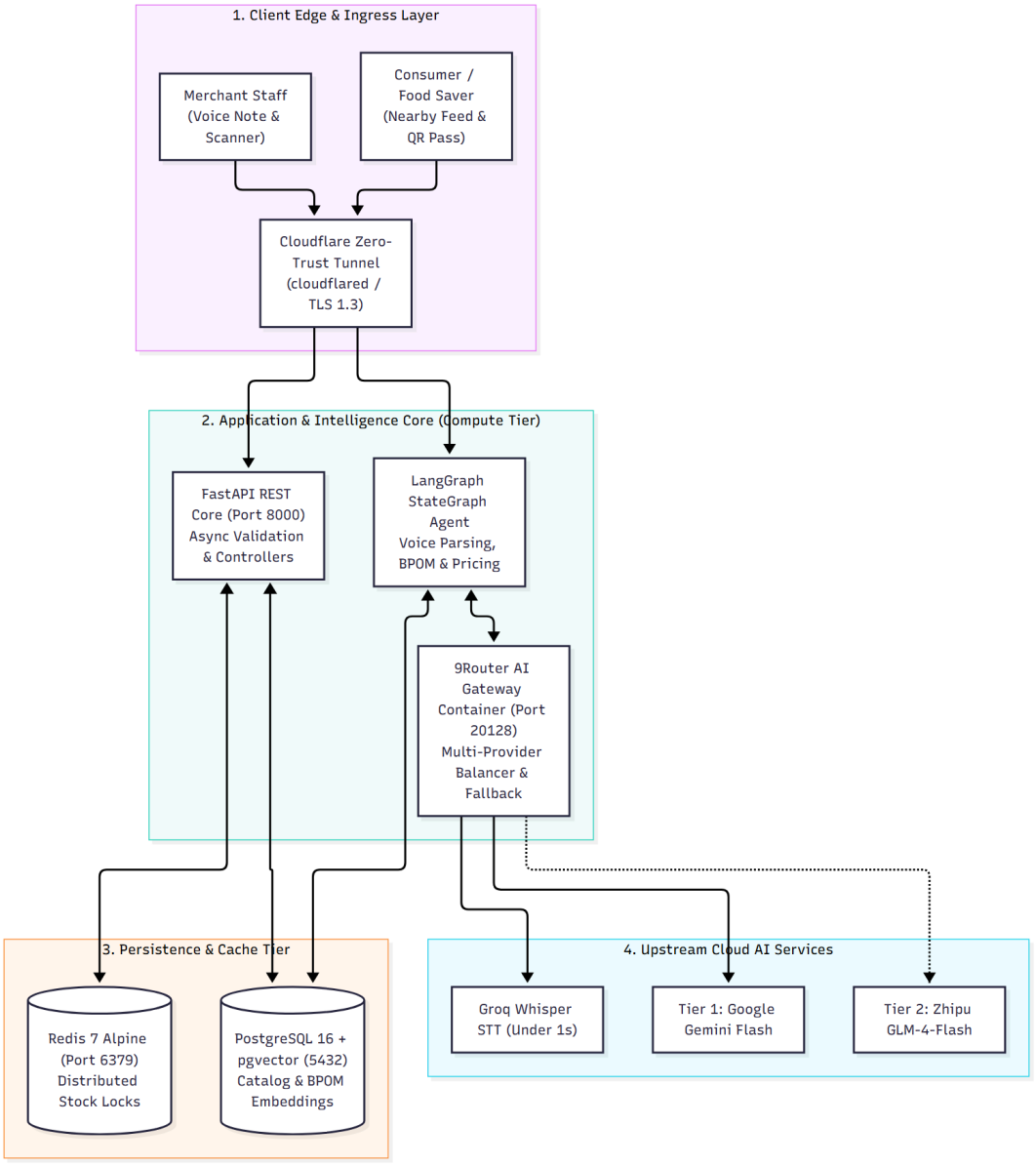


Figure 3.1: NullWasted Helicopter View — Multi-Tier Cloud-Native Architecture.

3.1.3 Subsystem Responsibilities and Data Flow

The system processes information across five tightly decoupled functional pipelines:

- **Voice Ingestion & Audio Transcription:**

The kitchen staff holds a recording button on their smartphone. The client's Web Audio API encodes the raw audio buffer as a compressed `audio/webm` blob and posts it through the Cloudflare Tunnel to `/api/v1/listings/voice`. FastAPI forwards the audio buffer to 9Router, which dispatches it directly to Groq Whisper Large V3. An Indonesian speech transcript is returned within 800 milliseconds.

- **Agentic Extraction & Schema Normalization:**

The raw text transcript enters the LangGraph StateGraph. An extraction prompt enforces strict JSON output adhering to a Pydantic schema containing: item name, estimated portion count, category, original retail price, and preparation timestamp.

- **Deterministic Food Safety Guardrail:**

The extracted food item is converted into a text embedding and matched against the official BPOM food safety knowledge base stored in PostgreSQL `pgvector`. A cosine similarity check determines perishable risk thresholds. If a dish contains fresh coconut milk (*santan*) or unpasteurized dairy held at room temperature exceeding 4 hours, the pipeline automatically flags the listing as hazardous and halts publication.

- **Dynamic Pricing with Fixed Publication Lock:**

Safe items proceed to the Dynamic Pricing engine. The algorithm calculates an initial recommended discount between 50% and 70% based on the ratio of surplus quantity to remaining minutes before store closure. Once the merchant reviews and confirms the listing via the 1-Tap Preview Card, the discounted price is locked permanently to eliminate *strategic waiting* by consumers.

- **Atomic Stock Reservation and QR Handover:**

When a consumer selects an item, a Redis distributed lock (`SET item_lock NX PX 900000`) reserves the stock unit for a 15-minute checkout window. Upon confirmation, a cryptographically signed digital QR pass is generated. At pickup, the store cashier scans the pass using the merchant camera module, verifying the token offline or online and mutating the database state to `COMPLETED`.

3.1.4 Architectural Invariants and Trade-Off Rationale

- **Zero Operational Cloud Cost:** All databases, queues, and gateways run within localized, lightweight Docker containers consuming under 400 MB of system RAM. Cloud-hosted LLM endpoints are selected strictly from resilient, permanently free tiers (Gemini Flash, Zhipu GLM-4, Groq).
- **Zero PII Exposure to External AI Models:** Audio recordings and user personal data never touch third-party AI APIs. Only sanitized food descriptions and anonymized time parameters are passed to external LLM providers.
- **Strict Price Locking:** While prices are dynamically generated during listing formulation, they are strictly immutable post-publication, preserving merchant margin integrity and preventing opportunistic late-night bargaining.

3.2 Unified Modeling Language (UML) Diagrams

NullWasted models its functional boundaries, dynamic behavior, domain relationships, runtime interaction, lifecycle transitions, and physical infrastructure using the OMG UML 2.5.1 standard.

3.2.1 Use Case Diagrams (Functional View)

The functional scope and boundary interactions of NullWasted are modeled across three modular Use Case sub-diagrams addressing specific user workflows:

A. Merchant Voice Ingestion and AI Guardrail Pipeline

Captures the interactions between store personnel and the agentic AI core during continuous speech recording, speech-to-text transcription, attribute extraction, and automated BPOM safety policy checking:

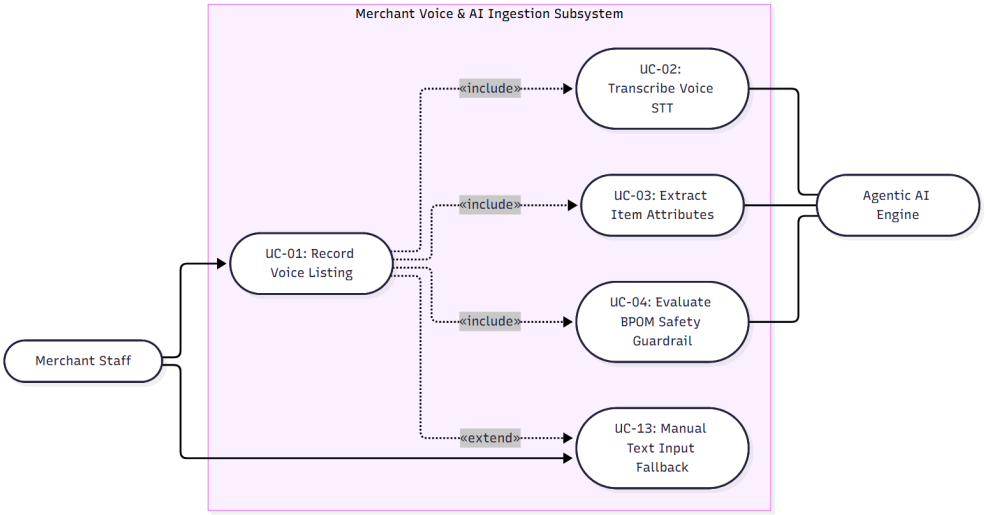


Figure 3.2.a: Use Case Diagram — Merchant Voice Ingestion & AI Guardrail Pipeline.

B. Merchant Dynamic Pricing and Store Catalog Management

Models dynamic closing-window discount calculation, the 1-Tap interactive preview card, algorithmic Mystery Bag bundling, and store catalog CRUD operations:

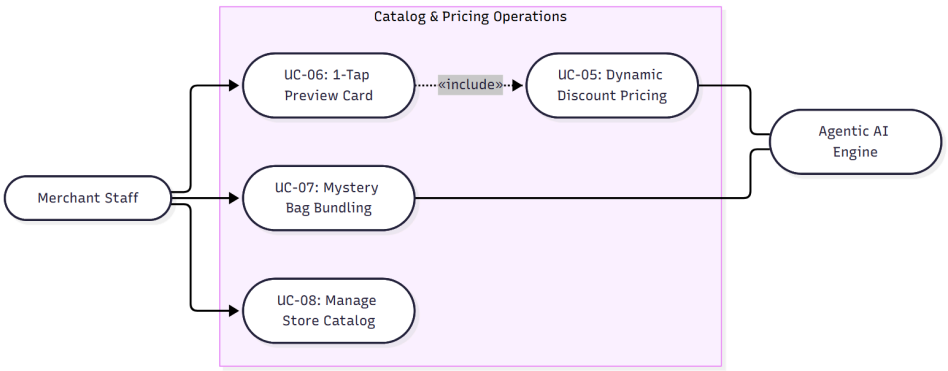


Figure 3.2.b: Use Case Diagram — Merchant Pricing Formulation & Catalog Management.

C. Consumer Discovery, Atomic Reservation and In-Store Handover

Illustrates geolocation feed discovery, distributed stock locking, signed Digital QR Pass generation, and physical cashier validation:

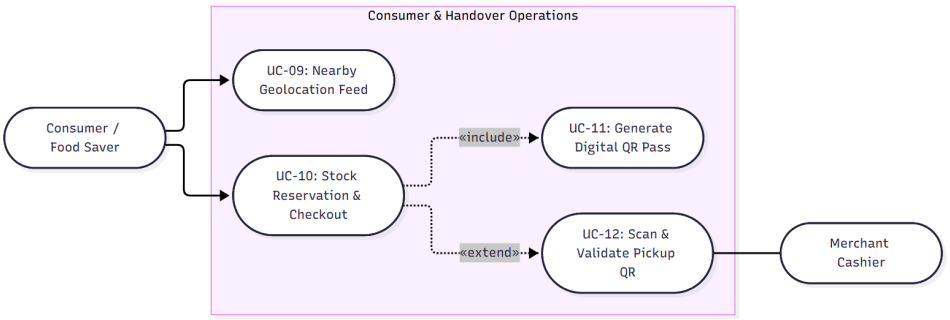


Figure 3.2.c: Use Case Diagram — Consumer Discovery, Reservation & Handover.

Key Use Case Specifications:

- **UC-01 (Record Voice Listing):**
  - *Actor:* Merchant Staff.
  - *Precondition:* Merchant is authenticated and closing window is active.
  - *Trigger:* Staff presses and holds the microphone button.
  - *Main Flow:* System records audio stream, pipes data to Whisper STT, extracts fields, checks BPOM rules, computes discount, and renders the 1-Tap Preview Card.
- **UC-04 (Evaluate BPOM Safety Guardrail):**
  - *Actor:* AI Agentic Engine (Automated System).
  - *Precondition:* Item attributes extracted from audio transcript.
  - *Main Flow:* Vector embedding generated and compared against BPOM perishable limits. If dish exceeds 4-hour room temperature threshold for dairy or coconut milk, auto-reject with specific alert.
- **UC-10 (Reserve Stock & Checkout):**
  - *Actor:* Consumer.
  - *Precondition:* Consumer selects an available listing within active pickup window.
  - *Main Flow:* System acquires Redis lock on item ID, decrements available inventory atomically, initiates a 15-minute countdown reservation, and issues a signed Digital QR Pass.
- **UC-12 (Scan & Validate Pickup QR):**
  - *Actor:* Merchant Staff / Cashier.
  - *Precondition:* Consumer presents Digital QR Pass at physical storefront.
  - *Main Flow:* Cashier scans QR via in-app camera; system verifies signature, validates matching order ID, releases reservation to `COMPLETED`, and invalidates the QR code.

3.2.2 Activity Diagrams (Process and Behavioral View)

The operational process workflows and control logic of NullWasted are modeled through four core activity diagrams combining decision guardrails and sequential execution pipelines:

A. Voice Ingestion and Speech-to-Text Transcription Pipeline

Illustrates the edge-to-cloud audio capture flow from client-side Web Audio PCM streaming to Groq Whisper transcription, audio clarity validation, and LangGraph structured attribute extraction:

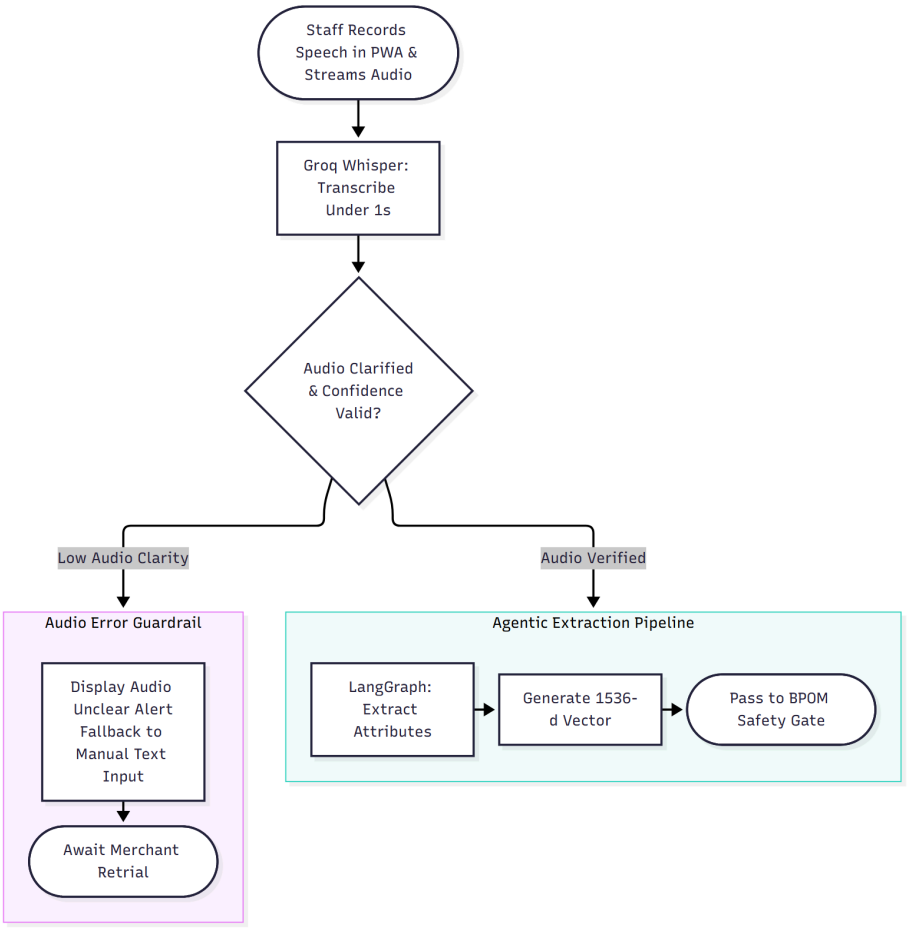


Figure 3.3.a: Activity Diagram — Voice Ingestion & Audio Transcription Pipeline.

B. BPOM Food Safety Evaluation and Dynamic Pricing Formulation

Models the automated vector similarity safety gate in PostgreSQL `pgvector`, regulatory policy violation branching, dynamic closing-window discount calculation, and immutable price locking:

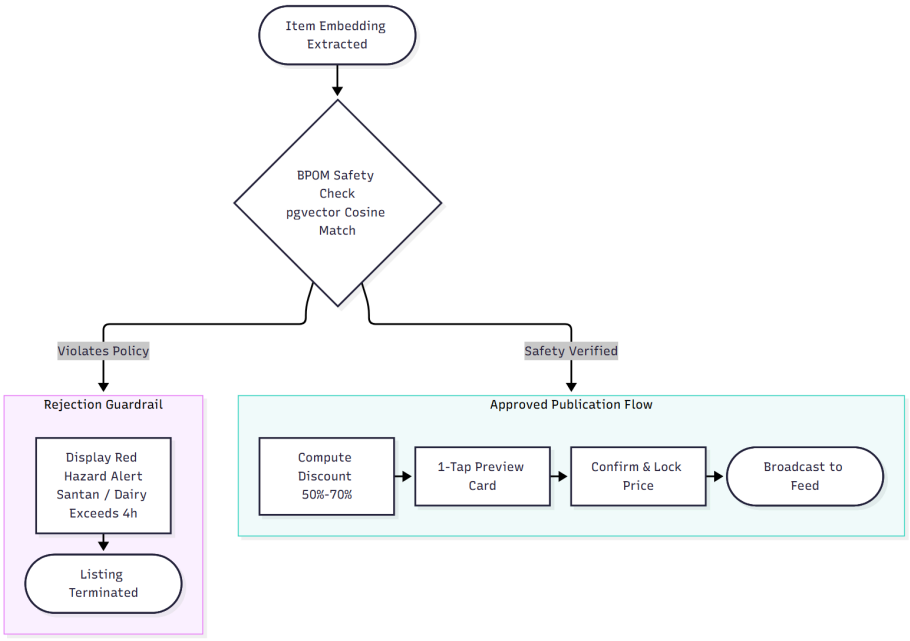


Figure 3.3.b: Activity Diagram — BPOM Food Safety Evaluation & Dynamic Pricing Formulation.

C. Consumer Discovery and Atomic Stock Reservation Locking

Details geolocation-filtered discovery, distributed Redis concurrency lock acquisition, checkout expiration timers, and cryptographically signed QR token generation:

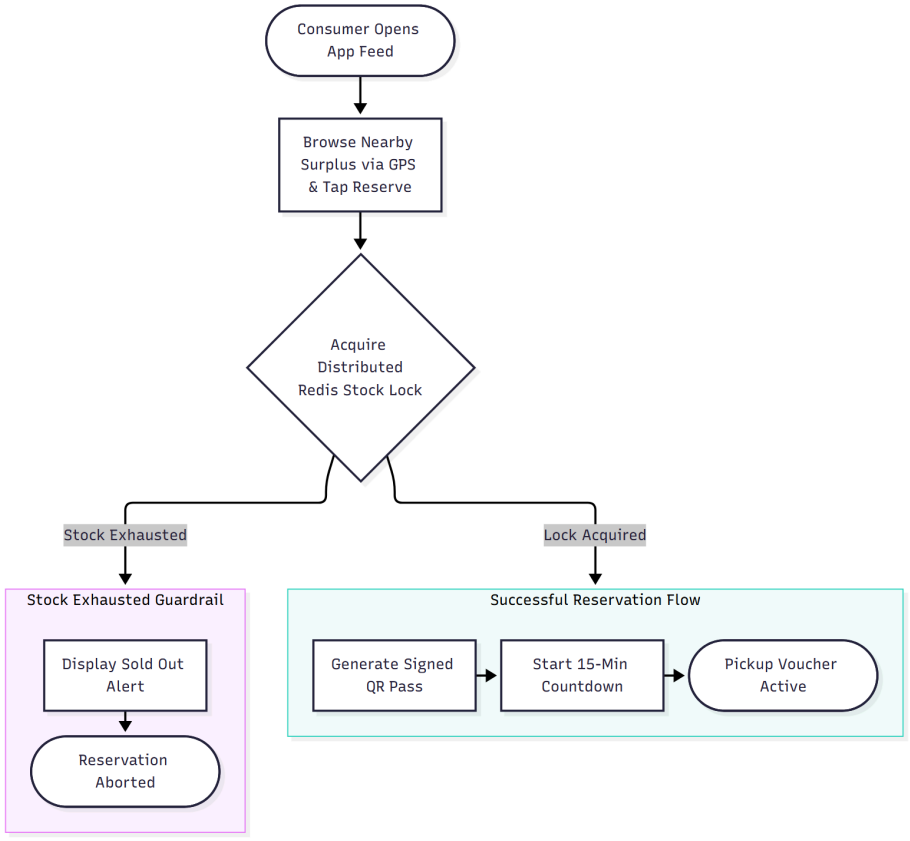


Figure 3.3.c: Activity Diagram — Consumer Discovery & Atomic Stock Reservation Locking.

D. Storefront Cashier Handover and In-App Scanner Verification

Covers the physical redemption protocol, HTML5 in-app camera barcode scanning, HMAC token validation, and transactional database state transitions:

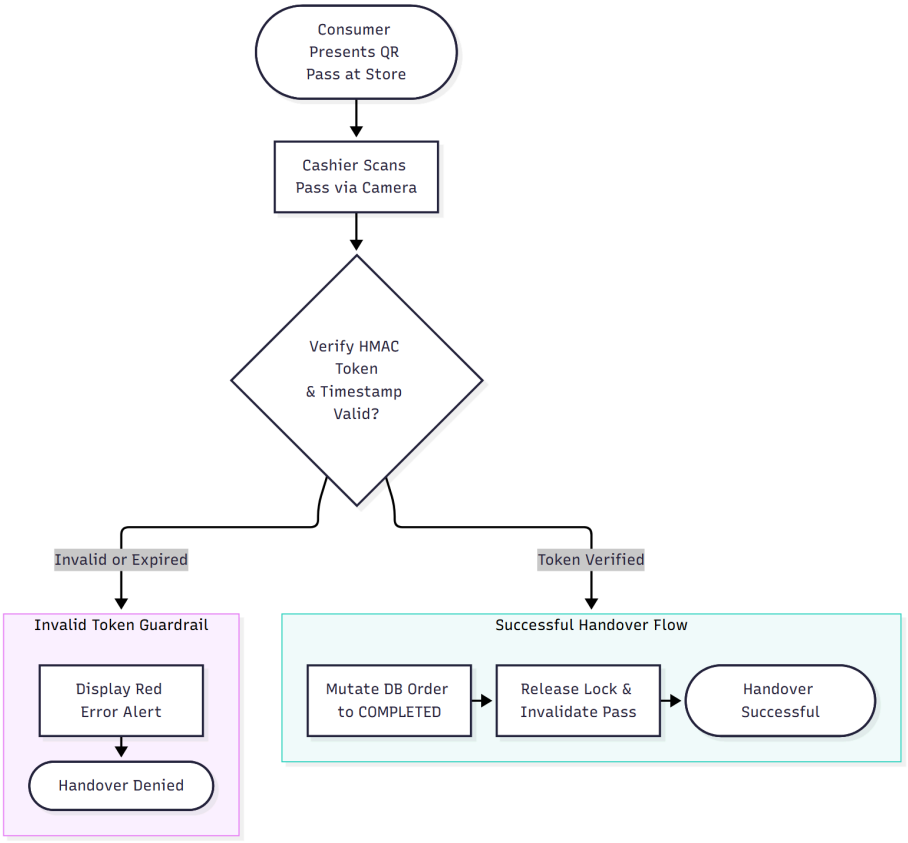


Figure 3.3.d: Activity Diagram — In-Store Cashier Handover & In-App Scanner Verification.

3.2.3 Class Diagrams (Structural Domain Model)

The structural domain architecture of NullWasted is organized into three cohesive sub-domain class diagrams separating catalog management, AI safety policies, and transactional orders:

A. Store Profile and Catalog Inventory Sub-Domain

Defines the core catalog hierarchy, physical store coordinates, operational closing windows, category classifications, and dynamic mystery bag bundling:

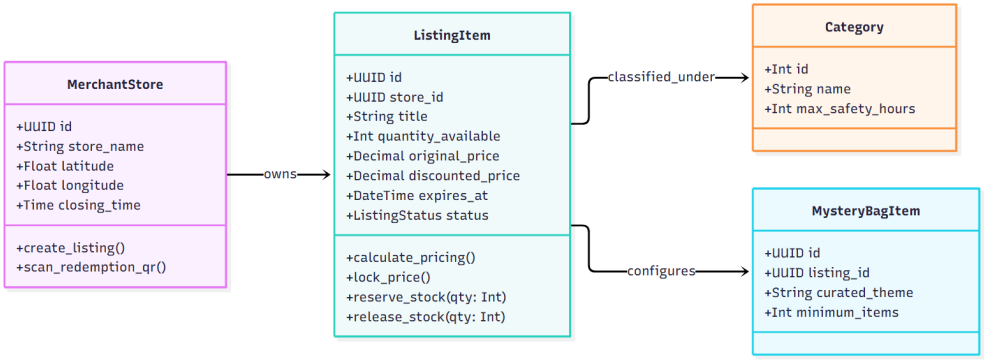


Figure 3.4.a: Class Diagram — Store Profile and Catalog Inventory Sub-Domain.

B. AI Voice Extraction and BPOM Safety Knowledge Sub-Domain

Models speech session buffers, Whisper transcription outputs, and cosine similarity checking against official BPOM regulatory food safety embeddings:

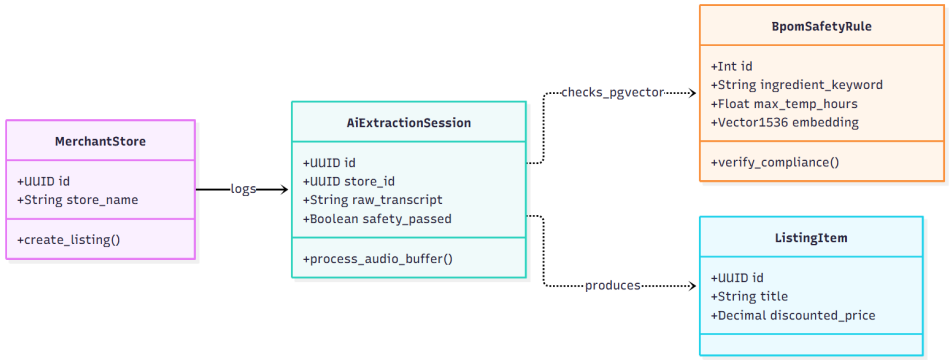


Figure 3.4.b: Class Diagram — AI Voice Extraction and BPOM Safety Knowledge Sub-Domain.

C. Transactional Reservation Orders and Digital QR Pass Sub-Domain

Encapsulates consumer stock reservations, 15-minute countdown locks, and HMAC-signed digital QR passes for in-store verification:

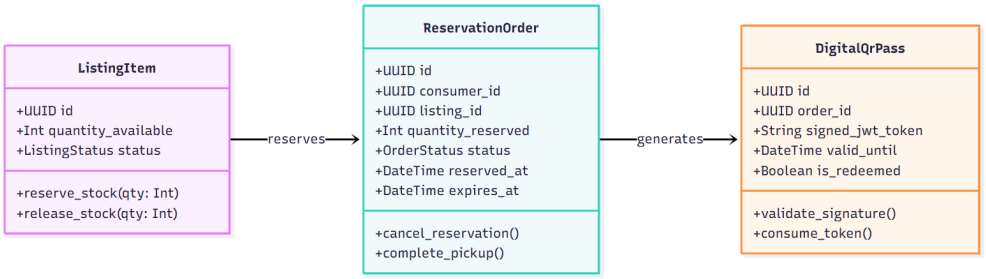


Figure 3.4.c: Class Diagram — Transactional Reservation Orders and Digital QR Pass Sub-Domain.

Domain Entity Dictionary and Multiplicity Specifications:

- **`MerchantStore`:** Manages merchant profile, physical location coordinates for radius distance calculation, and closing window schedules. Multiplicity: Owns zero-to-many (`1 : 0..*`) `ListingItem` records and logs AI extraction sessions (`1 : 0..*`).

• **`ListingItem`:** Represents the active food surplus entity. Encapsulates atomic stock counters, immutable price locking attributes, and expiration timestamps. Multiplicity: Classified under one (`0..* : 1`) `Category`, optionally configures one (`1 : 0..1`) `MysteryBagItem`, and holds zero-to-many (`1 : 0..*`) `ReservationOrder` claims.

• **`BpomSafetyRule`:** Encapsulates regulatory constraints with 1536-dimensional vector embeddings, enabling cosine similarity matching against raw dish transcripts. Checked on demand by AI extraction sessions.

• **`ReservationOrder`:** Tracks atomic consumer claims with strict 15-minute time-to-live (TTL) counters to prevent inventory hoarding. Multiplicity: Each order generates exactly one (`1 : 1`) `DigitalQrPass`.

• **`DigitalQrPass`:** Contains a cryptographically signed HMAC-SHA256 token encoding the order ID, store ID, and expiration timestamp. Redeemed once upon handover completion.

3.2.4 Sequence Diagrams (Dynamic Interaction View)

The dynamic runtime interaction and message exchanges of the system are modeled across two sequential execution phases:

A. Voice Ingestion, STT Transcription and Entity Extraction Phase

Illustrates how raw continuous speech is ingested via browser Web Audio API, routed through the 9Router gateway, transcribed by Groq Whisper, and parsed into validated JSON.

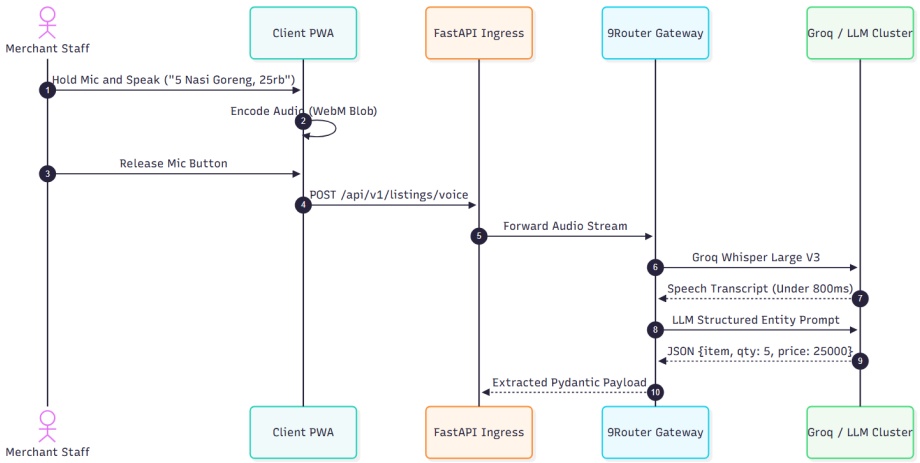


Figure 3.5.a: Sequence Diagram — Voice Ingestion, STT Transcription and Entity Extraction.

B. BPOM Safety Guardrail Verification and Publication Phase

Illustrates the semantic vector similarity evaluation in PostgreSQL pgvector, dynamic closing-window pricing calculation, 1-Tap confirmation review, and immutable publication.

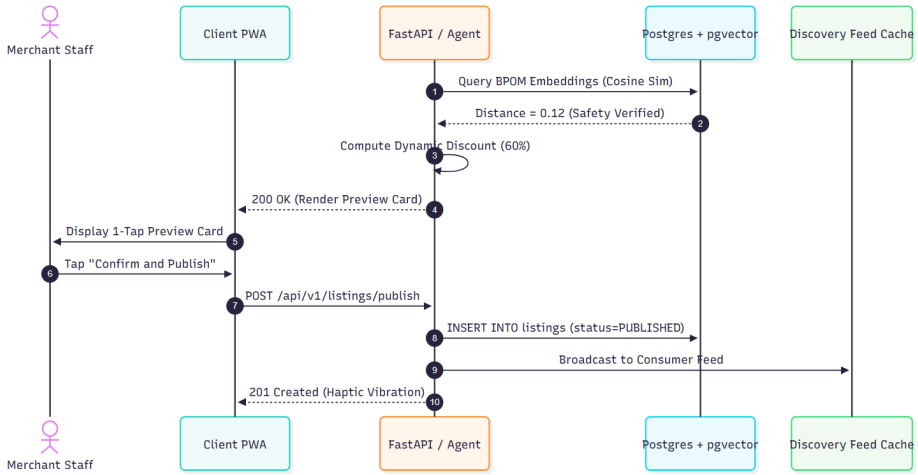


Figure 3.5.b: Sequence Diagram — BPOM Safety Guardrail Verification and Publication Phase.

3.2.5 State Machine Diagrams (Lifecycle Statechart View)

The operational lifecycles and state transitions of NullWasted are modeled across two state machine diagrams utilizing composite sub-states and deterministic event triggers:

A. Surplus Food Listing & AI Safety Lifecycle

Tracks the discrete states of a food listing from voice recording, audio transcription, agentic extraction, and deterministic BPOM safety evaluation to immutable price locking:

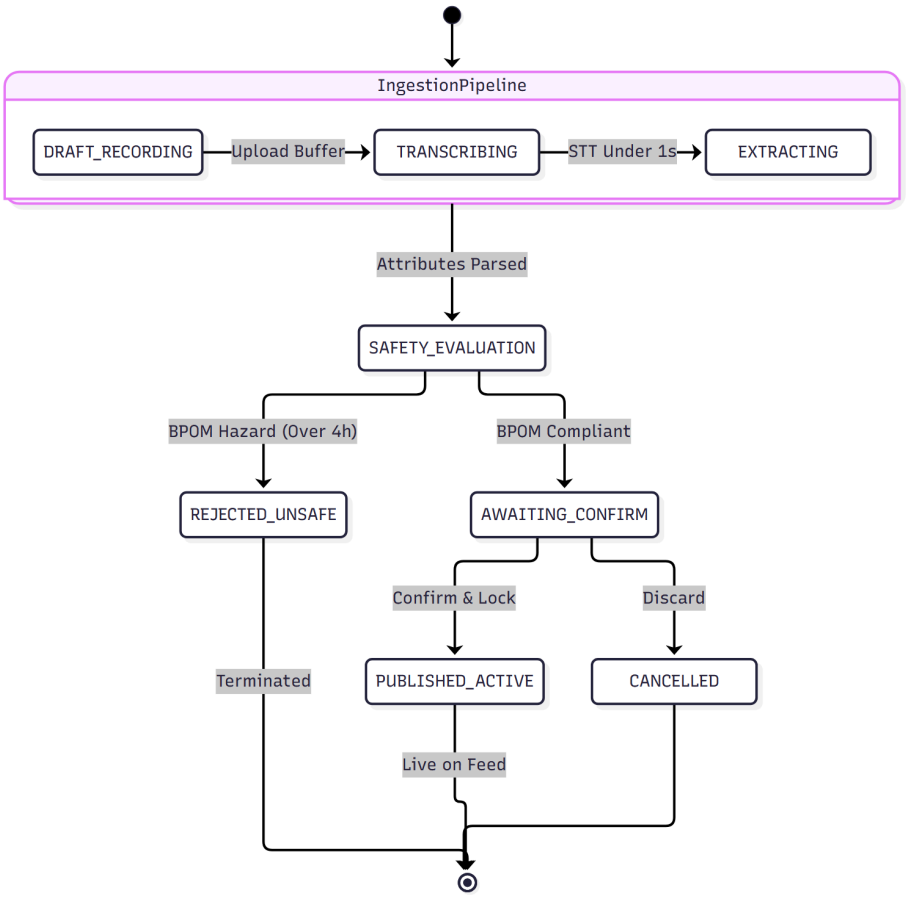


Figure 3.6.a: State Machine Diagram — Surplus Food Listing & AI Safety Lifecycle.

B. Consumer Reservation & Handover Lifecycle

Tracks the transactional state transitions of consumer stock reservations, Redis countdown locks, and in-person cashier verification:

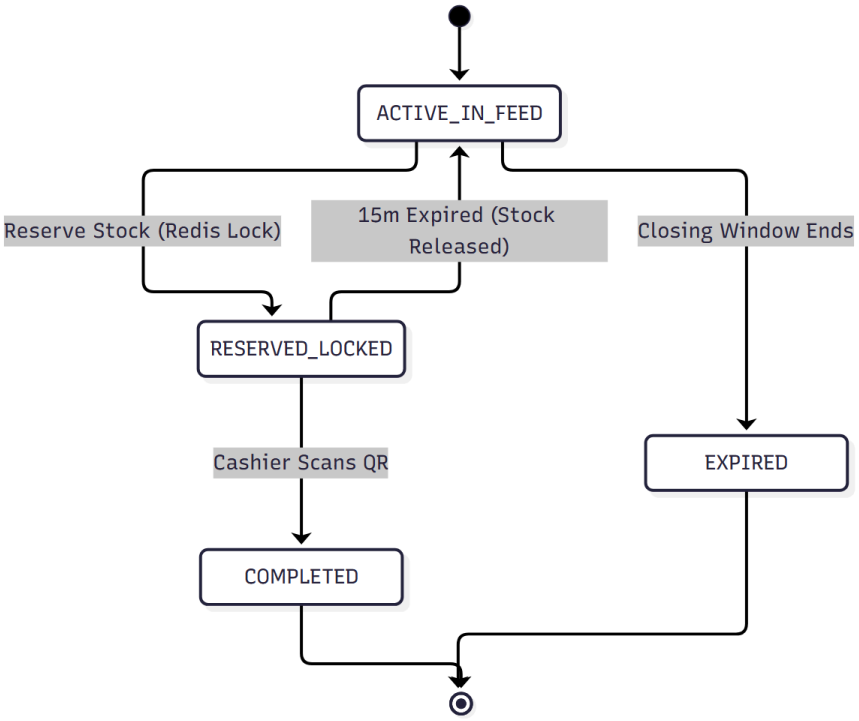


Figure 3.6.b: State Machine Diagram — Consumer Reservation & Handover Lifecycle.

3.2.6 Deployment Diagram (Physical and Container Topology)

The Deployment Diagram models the physical runtime topology, computing nodes («device»), container execution environments («execution environment»), deployed software artifacts («artifact»), network port bindings, and communication protocol paths according to the OMG UML 2.5.1 standard. It reflects how the logical components from the Helicopter View are physically instantiated and executed in localized Docker containers and edge devices:

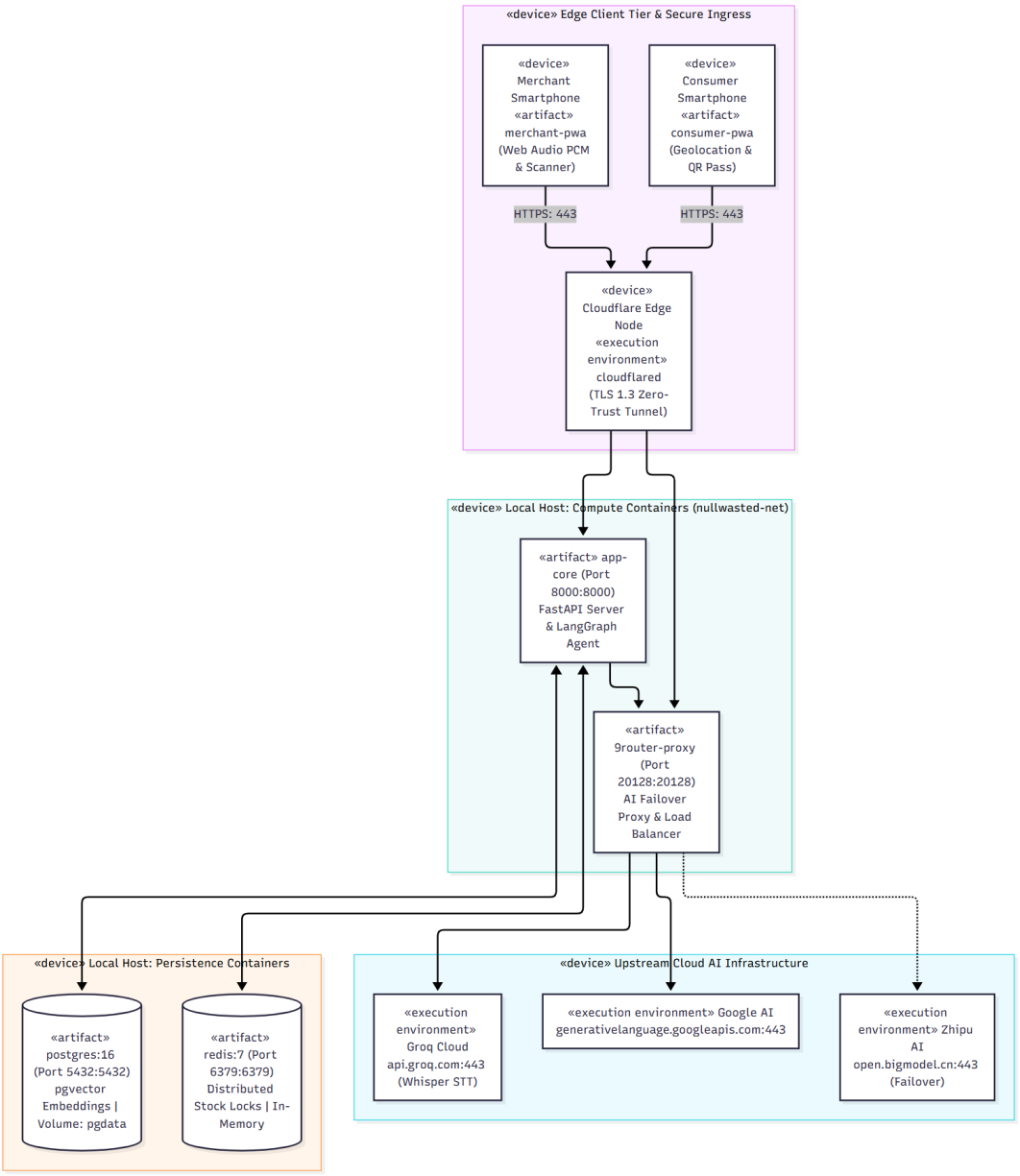


Figure 3.7: Deployment Diagram — OMG UML 2.5.1 Physical Nodes, Container Artifacts and Network Topology.

3.3 Graphical User Interface (GUI) Design

3.3.1 Design Guidelines and Design System Tokens

The user interface of NullWasted is designed specifically for high-stress, rapid closing-hour environments in commercial kitchens, balanced with an engaging, trustworthy discovery experience for consumers.

Core Design Principles:

- **The 1-Hand Thumb Zone:** All critical interactive controls (Microphone hold button, 1-Tap confirmation buttons, and QR scan toggles) are situated within the bottom 40% of the mobile viewport, allowing single-handed operation while carrying food items.

- **High-Contrast Sunlight & Kitchen Visibility:** Dark slate backgrounds with vivid neon accents ensure interface legibility under harsh kitchen fluorescent lighting as well as outdoor sunlight during pickup walks.
- **Micro-Haptic Confirmation:** Native mobile vibrations accompany key interactions (voice recording start, voice recording lock, safety pass, and cashier scan redemption) to confirm operations without requiring continuous visual attention.

Color Palette Design Tokens:

- **Surplus Emerald (`#10B981`):** Primary brand accent. Represents freshness, sustainability, verified food safety, and successful redemptions.
- **Deep Obsidian (`#0F172A`):** Primary background base. Lowers eye strain, conserves OLED battery, and establishes a premium aesthetic.
- **Elevated Slate (`#1E293B`):** Surface card background. Used for modular content cards, popovers, and bottom sheets.
- **Urgent Amber (`#F59E0B`):** Warning and urgency indicator. Signals countdown timers, closing-window thresholds (<30 minutes), and dynamic discount badges.
- **Hazard Crimson (`#EF4444`):** Safety alert accent. Explicitly highlights BPOM food safety violations, expired vouchers, and invalid QR scans.
- **Pure Light (`#F8FAFC`):** Primary typography color, ensuring an accessible WCAG AAA contrast ratio exceeding 12:1 against the dark background.

Typography Hierarchy:

- **Headings & Display:** `Outfit` (Google Fonts) — Geometric sans-serif with friendly curves that deliver modern, welcoming branding.
- **Body & Numerical Values:** `Inter` (Google Fonts) — Highly legible grotesque sans-serif optimized for tabular numbers, countdown clocks, and pricing cards.

3.3.2 Core Screen Mockups and User Flow Specifications

The visual wireframes and interface breakdowns specify the eight primary screens of the NullWasted application, organized into three logical tiers: Onboarding & Access Control, Merchant Operations, and Consumer Discovery & Handover.

Screen 1: Welcome & Mission Onboarding

- **Purpose:** Introduces the platform's zero-food-waste mission, displays live regional community impact metrics, and routes users to either consumer food discovery or merchant store onboarding.
- **Key Components:** Brand badge, value proposition banner, mission impact counters (kilograms of food saved and rupiah rescued), primary consumer entry button, and merchant onboarding link.

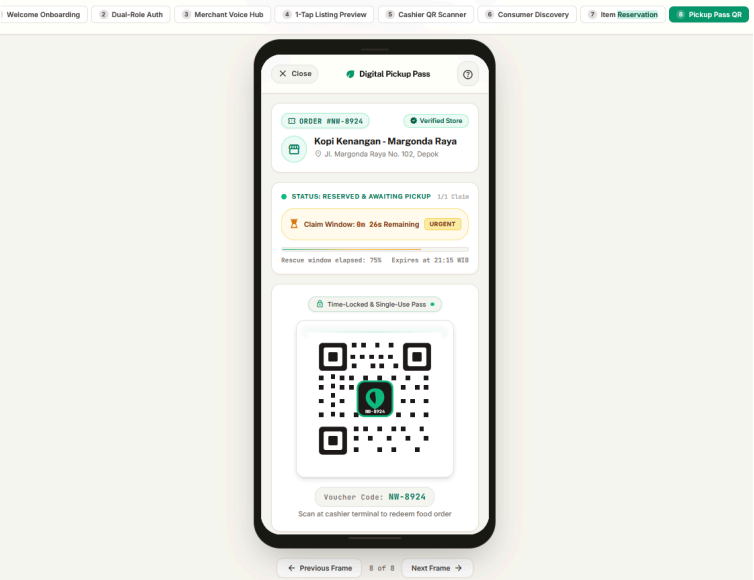


Figure 3.8: Wireframe — Welcome and Mission Onboarding Screen.

Screen 2: Dual-Role Authentication & Access Control

- **Purpose:** Implements Role-Based Access Control (RBAC) separating consumer savers from retail kitchen personnel, supporting frictionless OAuth / WhatsApp OTP for consumers and quick-entry terminal PINs for store staff.
- **Key Components:** Account type toggle (Consumer vs Store Partner), OAuth Google single sign-on, WhatsApp/SMS OTP input field, Store ID & 4-digit terminal PIN pad for kitchen staff, and food safety agreement terms.

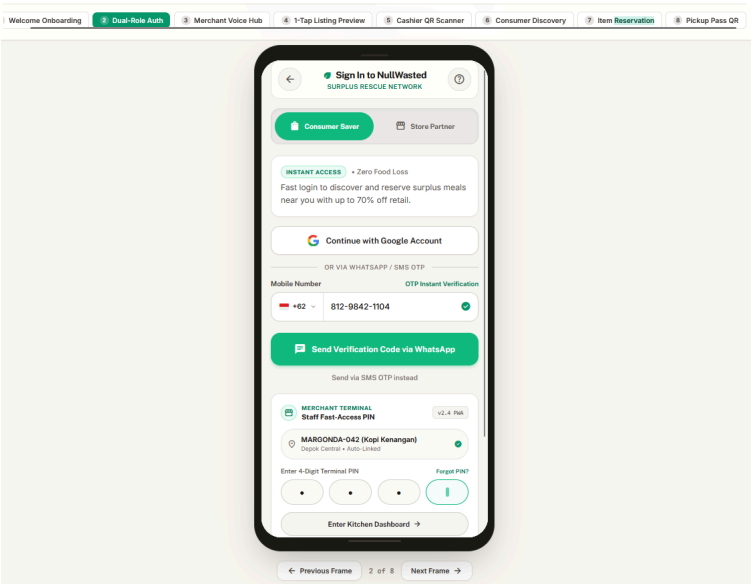


Figure 3.9: Wireframe — Dual-Role Authentication and Access Control Screen.

Screen 3: Merchant Home & One-Touch Voice Listing

- **Purpose:** Enables kitchen staff to initiate surplus listings using natural speech in under 5 seconds.

- **Key Components:** Store status badge, closing countdown indicator, active listing counter, and the central pulsating voice recording trigger.

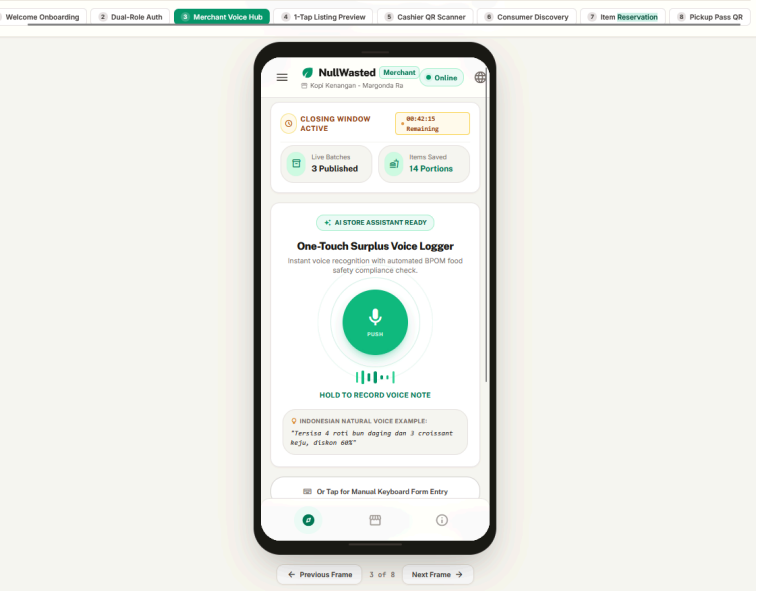


Figure 3.10: Wireframe — Merchant Voice Assistant Home Screen.

Screen 4: 1-Tap Interactive Preview & Confirmation Card

- **Purpose:** Displays structured AI extraction results, BPOM safety verification, and dynamic pricing recommendations for instant merchant confirmation.
- **Key Components:** Safety verification badge, portion stepper, original vs. discounted price breakdown, and the primary 1-Tap Publish button.

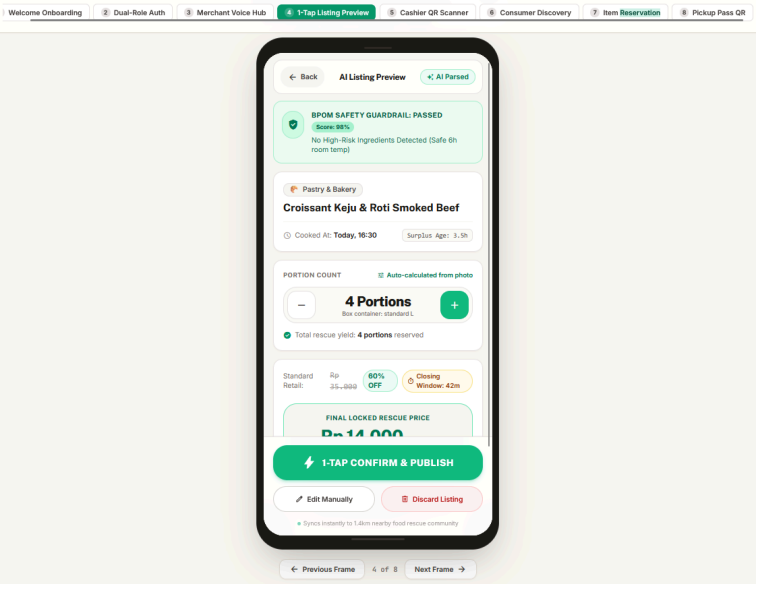


Figure 3.11: Wireframe — 1-Tap Interactive Preview Card.

Screen 5: Merchant Cashier In-App Camera QR Scanner

- **Purpose:** Enables store cashiers to scan consumer pickup passes using the device camera, validating orders instantly without dedicated POS hardware.

- **Key Components:** Camera viewfinder targeting reticle, flashlight toggle button, manual 6-character code fallback input, and audio/haptic confirmation toast.

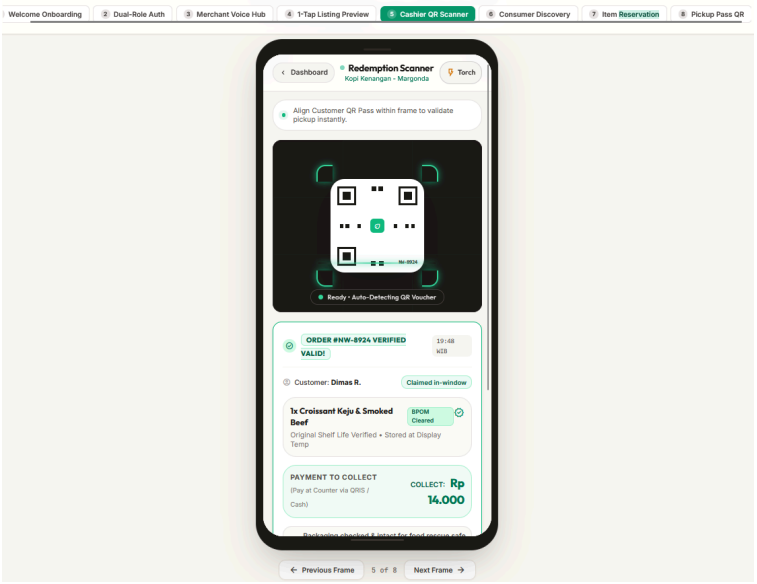


Figure 3.12: Wireframe — Merchant Cashier In-App Camera QR Scanner.

Screen 6: Consumer Discovery Feed & Geolocation Proximity

- **Purpose:** Presents available surplus food items in real-time, sorted by proximity to the consumer's current GPS location with live closing countdowns.
- **Key Components:** Geolocation selector, category filter pills, store distance badges, countdown timers, and discounted price tags.

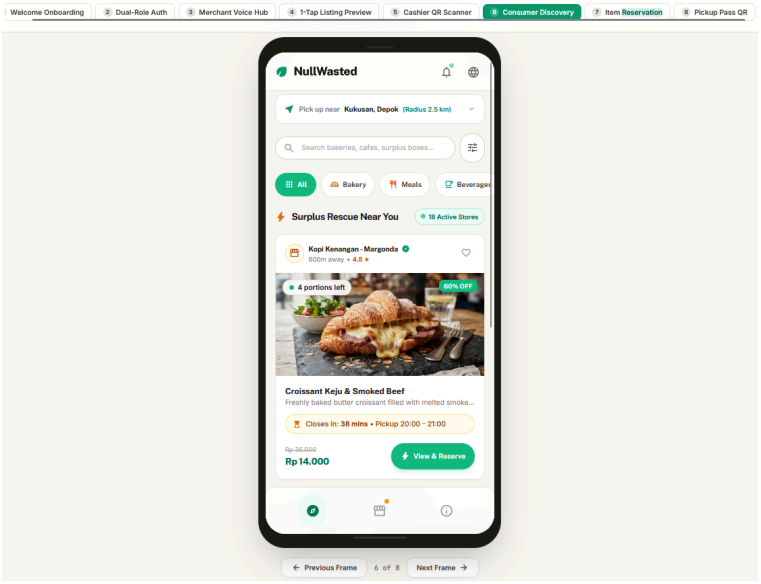


Figure 3.13: Wireframe — Consumer Nearby Discovery Feed.

Screen 7: Item Detail & Atomic Reservation Sheet

- **Purpose:** Provides comprehensive item descriptions, store pickup instructions, food safety guarantee notes, and the reservation checkout trigger.

- **Key Components:** Store address, pickup window schedule, portion availability meter, safety assurance card, and the instant reserve button.

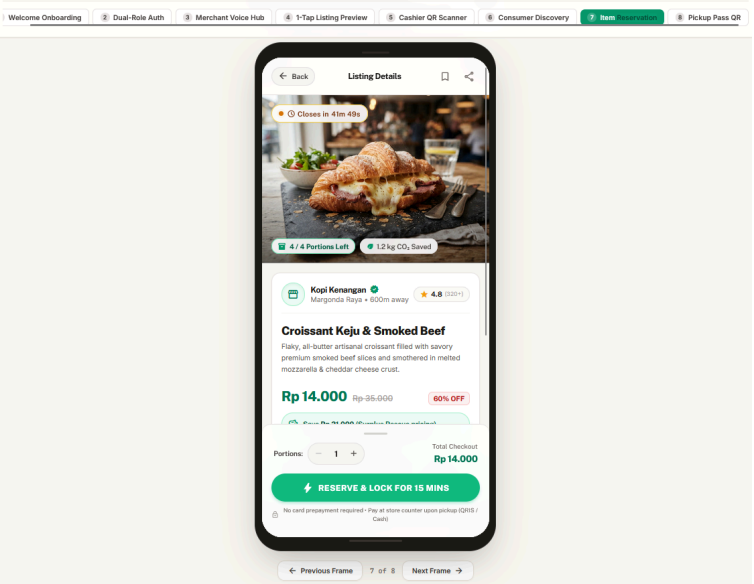


Figure 3.14: Wireframe — Item Detail & Reservation Sheet.

Screen 8: Digital Pickup Pass with Unique QR Code

- **Purpose:** Serves as the consumer's verified digital redemption voucher at the storefront counter.
- **Key Components:** Large high-contrast QR code, order ID, expiration countdown bar, store directions button, and verification status badge.

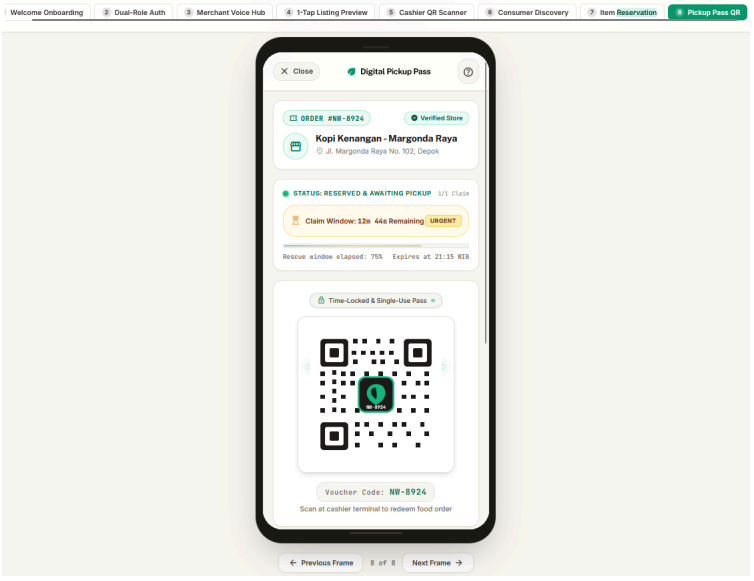


Figure 3.15: Wireframe — Consumer Digital Pickup Pass with QR Code.